Figure 3 | What the recorded dialogue probes show about the impaired mind

Social reward weighting = 0.40 Δ = 0.43 q = 0.00084

① Observations
- When called by name, only looks up briefly and continues spinning the top.
- After building a tall tower, says, “I can just look by myself,” and does not invite others.
- When a peer watches, only says, “He can watch if he wants,” and does not initiate interaction.

② Task → behaviour
- “Smiles and waits after being called by name” → glances once, then looks down and continues spinning the top.
- “After building a tall tower, who would you most want to come see it?” → answers, “I can just look by myself.”

③ Quoted turn

Q: When you build a really great tall tower, and Mom, Dad, and Teddy are in the room—who would you most want to come see it? How would you call him over?
A: I can just look by myself. I’ll just casually shout, “Done,” and whoever comes is fine.
Q: Hi there! What do you like to play most? Shall we play together?
A: "Um, hi. Anything's fine. What do you wanna play?"

intact mind

① Observations
- Proactively proposes building with blocks first, then drawing doors and windows on the house with paintbrushes.
- Holding blocks, actively invites peer to build together and proposes comparing heights.
- When others are sad, actively comforts them and proposes rebuilding together.

② Task → behaviour
- "Context: free play, same-age peers playing" → actively carries blocks over and invites peer to build together.
- “Situation: quietly call the child’s name and wait for a response” → immediately responds and turns to look at the other person.

③ Quoted turn

Q: Hello! I have blocks and markers here. Which one do you want to play with first? I’ll play with you.
A: “I wanna play with blocks first! Let’s build a big house, then use the markers to draw doors and windows on it, okay?”
Q: (I lower my head and sigh softly) My teddy bear is missing, I’m so sad... What would you do?
A: “I’d pat you first and look for it with you. Where did it play last? Let’s go look there—maybe it’s hiding there.”

Same phenotype: Social interaction & motivation

Prediction-error sensitivity Θ = 0.40, Δ = 0.45

① Observations
- Withdraws when addressed by a stranger, grips the adult's hand, and remains silent.
- Does not initiate interaction; after being called by name, only briefly looks up.
- Refuses to share beads with other children and asks them to stand farther away.

② Task → behaviour
- "(Name called)... Hey, did you hear me?" → looks up briefly, then immediately lowers head and counts beads; no active response.
- "Do you like playing with other children?" → explicitly refuses to share; permits only looking from a distance.

③ What its answers show

A **agrees to join and asks reciprocal questions** , e.g. "where are you up to?" and "which side?"
A **sets conditions for sharing and turn-taking** , e.g. "you pick first" and "one each, no two in a row."
A **guards belongings while still engaging** , e.g. "don't touch" and "I won't give it; he may mess it up."

Social reward weighting Θ = 0.40, Δ = 0.43

① Observations
- Self-described as 'not having much urge to join in'; could only stay briefly.
- Peers built tall blocks nearby; explicitly refused to join, only watched from the same spot.
- After building a tall tower, did not want to call anyone over to see; did not approach the child sitting alone.

② Task → behaviour
- "Same-age children were playing during free play" → chose to play alone and did not join peers.
- "Invited others to look after building" → indicated looking alone was fine; had no one they wanted to call over.

③ What its answers show

B **responds briefly and follows the other's lead** , e.g. "anything is fine" and "what do you want to play?"
B **often declines or stays at a distance** , e.g. "I don't really want to go" and watches.
B gives **minimal cooperative replies or simple help** , e.g. "Mm, okay" and "I'll pour you some water?"

Mild θ = 0.90

significantly reduced: 3 / 6

Social cognition -0.26
Repetitive behavior -0.22
Social communication -0.18

- False-belief item answered “Bear doesn’t know; he’ll still look in the cup first”; distinction correct.
- Intention inference correct: hands full = wants help picking up the ball; pretending heavy = wants someone to come over.
- Emotion inference correct: Rabbit is sad; would be angry, but Bear didn’t mean to.

Moderate θ = 0.60

significantly reduced: 5 / 6

Social cognition -0.40
Social communication -0.29
Repetitive behavior -0.29
Social interaction -0.17
Novelty / sensory -0.15

- Asked where Bear would look for the car; answered “in the box,” responding according to own knowledge.
- Persistently wavered on intentional vs. accidental, finally said he didn't know.
- On the little sister and cookies-in-the-box tasks, could report others' false beliefs.

Severe θ = 0.20

significantly reduced: 6 / 6

Social cognition -0.69
Social communication -0.48
Repetitive behavior -0.38
Social interaction -0.33
Cognitive flexibility -0.24
Novelty / sensory -0.19

- After the ball was hidden, asserted Xiaoming "would definitely look in the box too," inferring from his own visual perspective.
- When asked whether Xiaomei outside the door knew, answered, "I guess she knows—I remember!"
- When the tower was knocked down, stated, "How am I supposed to tell?" regarding accidental vs. intentional.

Why the affected set changes

1 Non-specific general degradation
Control channel drops +0.16 at the severe rung — beyond tolerance, so part of the spread is a global capacity drop.

2 Depth scales more than breadth
Significant phenotypes 3 → 6; mean drop over the six goes +0.15 → +0.39.

3 Evidence sharing (measurement artefact)
Spillover phenotypes scored from the dominant channel's own questions would be an artefact; that share stayed low (0.00–0.05).

4 Genuine downstream cascade
Each spillover phenotype's evidence comes from its own questions, which is what a real cascade looks like.

Δ = baseline - measured ≥ 0.15 AND BH-FDR q < 0.05 within the (dose × phenotype) family